Task 3- COMPLETE

Wednesday, November 13, 2024 2:54 PM

Hydraulic
Uses brake fluid, cylinders and friction to create pressure and makes brake pads stop wheels

Pros
Brake failure is unlikely as there is a direct connection between actuator and brake disc.
Good, consistent braking power
Smooth and precise braking (modulation and control
Brake pads are self-adjusting – less need for frequent manual adjustments

Cons
Pricey
Potential for fluid leaks (this is rare, only happens if system is damaged or not maintained well)
Requires maintenance to remove air bubbles – complex process, periodic.

Electromagnetic
Allows for frictionless braking using electromagnetism: when magnetic flux is in a direction perpendicular to the rotating wheel direction, and opposing force to wheel direction is produced.

Pros
Increased life span and reliability of brakes (few moving parts)
Rapid response times
Not ridiculously big
No maintenance cost (contactless so brake shoes don't have to be replaced periodically)
Lower running cost than other brakes as part of the energy is delivered to the supply

Cons
Generates significant heat during prolonged use
Reduced torque capacity (may restrict use in monorail where high braking force is required)

Mechanical:
A brake force is applied to brake pedal, force is transferred to the final brake drum/disc rotor via fulcrums, springs, cylindrical rods etc. – causing braking.

Pros
Easy to maintain – simple system + adjustments made easily
Cheaper than others

Cons
Provides less braking power
Brake pads wear down – maintenance required often for optimal braking performance
Powers emergency brake
Not very effective – cables stretch/wear out = replacement

In an air-brake system, compressed air forces a piston-driven brake shoe against the wheel. The brake shoes are made from a number of different substances, cast iron and synthetic materials.
In some modern systems, the braking force is not applied directly to the wheel. E.g. A disc brake system.

Basic Principles of Air Brake Systems

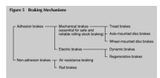
Automatic Air Brake System

Air compressors mounted every two to four cars supply compressed air to the air brakes. The air which is compressed to 700 to 900 kPa, is piped under the car floors to main air reservoirs. The air pressure is lowered to 490 kPa by a pressure regulator and the air is fed via the brake valve, brake pipes , and control valves to the auxiliary air reservoirs. When the compressed air in the brake pipes and auxiliary air reservoirs of each car is 490 kPa, the brakes are not activated. However, the activated brake valve cuts the flow of air from the pressure regulator, so the air pressure in the brake pipe falls. The fall in pressure is detected by the control valves on each car. The control valves then regulate the flow of compressed air from the auxiliary air reservoirs to the brake cylinders. The brake cylinders activate the basic brakes mechanisms to slow down and stop the car. The control valves regulate the flow of air from the auxiliary air reservoirs to the brake cylinders at a pressure that is proportional to the pressure drop in the brake pipes.

Straight Air Brake System

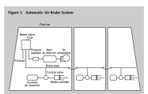
Unlike the automatic air brake system the straight air brake system does not have a control valve or auxiliary air reservoir in each car. Activation of the brake valve forces compressed air through the straight air pipe to the brake cylinders, activating the basic braking mechanism. However, since the straight air pipes do not contain compressed air during normal running conditions, the brakes would fail if cars became uncoupled. To avoid this problem, the straight air brake system can be used in conjunction with the automatic air brake system. It can also be avoided by running another pipe, called a main air reservoir pipe, from the first to the last car. The air pressure in this pipe acts like the compressed air in the brake pipes of the automatic air brake system. If the compressed air in this main air reservoir pipe fails, or if it leaks from air pipes or from air hoses between cars, etc., the pressure drop is detected and the brakes are applied automatically. For example in shinkansen, the brakes are applied automatically if the air pressure in the piping falls to 600 kPa.

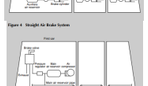
Braking Mechanisms

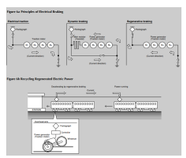


Government regulations specify braking distance and deceleration rates to ensure safe operation of a rolling stock. In Japan trains are at maximum speed of 130km/h and are required to be able to come to a stop within 800m. Trains are also equipped with failsafe mechanisms to ensure quick stops. When a braking force is applied to stop the cars, the force must be transmitted to the rails. The braking force can be transmitted either through adhesion, which makes use of friction at the point where the wheels touch the rails, or through non adhesion brakes. Non-adhesion methods do not use friction at the point where the wheels touch the rails and include mounting panels on cars to increase air resistance, or the direct application of pressure from the car to the rail, using shoes. This letter technique involves a device called

Control Brake (https://www.railway-technology.com/) or electrical brakes, or mechanical brakes.
A braking system used by electric trains is electrical dynamic braking that converts the motor into a braking generator dissipating the kinetic energy as heat. Regenerative braking uses the generated electricity instead of dissipating it as heat, and is becoming more common due to its ability to save energy. Although the traction motor drives and accelerates the train, during braking, it acts as an electric generator instead, forming part of a circuit that consists of a main resistor (rheostat), armatures and a field system. Electricity flows through the circuit and is consumed by the main resistor, which converts the kinetic energy of the train into heat and thereby acts as a brake.
Regenerative braking uses the same type of circuit, but the electricity generated by braking is not consumed by the main resistor. Instead it is transmitted to the overhead wire. The flow of this electricity is controlled by a controller under the pantograph that opens and closes with spill-second timing. Electrical brake systems are economical because they do not use friction elements, unlike mechanical brake systems. The regenerative braking system is even more economical because the electricity regenerated from the trains kinetic energy is transmitted to the overhead wire, and becomes available to power other rolling stock. The problem with electrical brake systems is that they occasionally







Lengthening time between air escape reservoirs.

1.3SIGNIFICANCES AND SCOPE

The following are the significances:

1. Electromagnetic brakes satisfy all the energy requirements of braking without the use of friction. They have better heat dissipation capacity and avoid problems that friction brakes face in use.
2. They can also be used as a supplementary retardation equipment in addition to the regular friction brakes on heavy vehicles.
3. These brakes' component cost is less so these brakes are cheap.
4. They can be used as an alternative method for the future crisis of the crude oils.

1.4Limitations

The following are the limitations:

1. The installation of an electromagnetic brake is difficult since there is not enough space between the gearbox and rear axle.
2. It cannot use grease or oil.
3. EM brakes are good at slowing things down, not completely stopping them.

1. Braking energy equal to a mass lifted

$$E = Mg = Ph$$

E = Energy (J)
M = Mass (kg)
g = gravitational acceleration (9.81 m/s²)
h = vertical distance the mass is lifted (m)

2. Braking force required to stop a mass lifted

$$F = Mg = Ma$$

F = Force (N)
M = Mass (kg)
a = deceleration (m/s²)
g = gravitational acceleration (9.81 m/s²)

3. Braking power required to lift a mass at a constant velocity

$$P = Fv = Mgv$$

P = Power (W)
F = Force (N)
v = velocity (m/s)
M = Mass (kg)
g = gravitational acceleration (9.81 m/s²)

4. Kinetic energy of a moving mass

$$E_k = \frac{1}{2}Mv^2$$

E_k = Kinetic energy (J)
M = Mass (kg)
v = velocity (m/s)

5. Braking power required to stop a moving mass in a specified time

$$P = \frac{E_k}{t} = \frac{\frac{1}{2}Mv^2}{t}$$

P = Power (W)
E_k = Kinetic energy (J)
t = time (s)
M = Mass (kg)
v = velocity (m/s)

1. REFERENCES

- Electric brakes (essential for safe and reliable rolling stock braking)
- Air resistance braking
- Rail brakes
- Anti-mounted disc brakes
- Wheel-mounted disc brakes
- Tread brakes (reducing pressure to increase adhesion)
- Potential to compress media (air) to braking
- Potential danger of this crushing out them (reducing pressure)
- Accurate change brake rate and intensity
- No oil pollution
- Easy to control (temperature and stabilizing, compressed air)
- Less expensive
- Simple (& easily combined with ordinary brakes)
- Can be utilized as a part of industry to stop or slow down machinery
- Revised rules.

2. CONCLUSION & FUTURE SCOPE

Electromagnetic brakes have numerous preliminary over mechanical brakes, including the ability to provide more rapid and efficient braking, fewer moving components, and reduced maintenance requirements. The use of electromagnetic brakes in various industries and applications is likely to increase in the future. The utilization of this can be dominated by advancing technology and growing demand for more efficient and reliable braking systems. The future scope of electromagnetic brakes is also influenced by the rise of electric vehicles (EVs) and the growing emphasis on sustainability and energy efficiency. These brakes are expected to play a key role in optimizing energy recovery and reducing carbon emissions. The development of more compact and lightweight electromagnetic braking systems with better heat dissipation and control capabilities will further enhance their adoption across various industries. Advancements in materials and manufacturing techniques will also be critical in ensuring the reliability and performance of these brakes.

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Electric brakes that use electromagnetic force force many advantages compared to mechanical brakes, which electromagnetic brakes do not use friction force but use magnetic force as braking effect. The magnetic field does not require regular replacement as happens with mechanical brakes which need to change brakings regularly, so that the electric brakes are perfect for use and combined with mechanical brakes to complement and improve the braking working system on the vehicle. A electromagnetic braking

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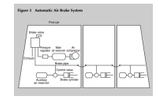
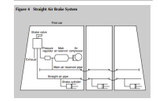
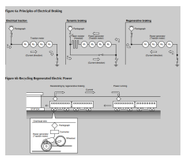
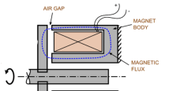
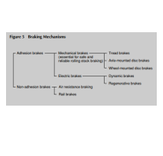
system that uses will be more environmentally friendly because the electric brakes do not produce pollution and the electric brakes also doesn't require a lot of power to do braking. Electric magnetic force is much easier to use in the application as a hybrid brake,so a vehicle because electromagnetic brakes can be used in the place of the friction brakes, and the electric brakes are reliable than the electromagnetic force generated has no side effects on the performance of mechanical brakes so electromagnetic brakes are right combined with mechanical brakes so that it becomes hybrid braking.

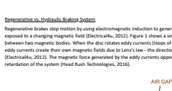


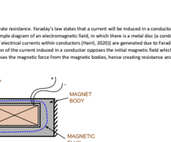


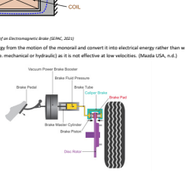












Regenerative vs. Hydraulic Braking System

Regenerative brakes stop motion by using electromagnetic induction to generate resistance. Faraday's law states that a current will be induced in a conductor which is exposed to a changing magnetic field (ElectricalAu, 2012). Figure 1 shows a simple diagram of an electromagnetic field, in which there is a metal disc (a conductor) situated between two magnetic bodies. When the disc rotates eddy currents (loops of electrical currents within conductors (Hairil, 2020)) are generated due to Faraday's Law. These eddy currents create their own magnetic fields due to Lenz's law - the direction of the current induced in a conductor opposes the initial magnetic field which produced it (ElectricalAu, 2012). The magnetic force generated by the eddy currents opposes the magnetic force from the magnetic bodies, hence creating resistance and causing retardation of the system (Head Rush Technologies, 2016).

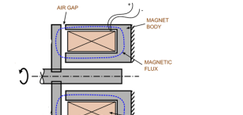


Figure 1: Diagram of an Electromagnetic Brake (SEPAC, 2021).

Regenerative braking allows electric or hybrid systems to capture kinetic energy from the motion of the monorail and convert it into electrical energy rather than waste energy (heat). However, it is often paired with a traditional braking system (i.e. mechanical or hydraulic) as it is not effective at low velocities. (Mazda USA, n.d.)

Figure 2: Diagram of a Hydraulic Brake (Quick, 2024)

Pascal's law states that when there is an increase in pressure at any point in a confined fluid, there is an equal increase at every other point in the container (Hodanbosi, 2014); this law allows hydraulic braking systems to be effective. Figure 2 shows a diagram of a hydraulic brake. When the brake pedal is pressed, brake fluid in the brake master cylinder becomes compressed. The fluid is then , however the brake fluid is incompressible, so the brake fluid becomes pressurised. The pressurization across the fluid is equal due to Pascal's laws transferred through the caliper brake to the brake piston which pushes the brake pad into contact with the wheel, generating friction which stops motion (ATC, 2022).

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Compare between regenerative, electromagnetic and hydraulic braking systems

COMPARISON OF BRAKING SYSTEMS

ELECTROMAGNETIC BRAKING SYSTEM PROS AND CONS

Pros

Offer rapid response times. (allows for quick engagement and disengagement) which is great for high speed machines.

Highly reliable and consistent as they have few moving parts

Enhanced safety as there's no hydraulic lines so they are less likely to fail due to line rupture.

Cons

Limited heat dissipation

Lower torque capacities so may be unsuitable for lower torque capacities

REGENERATIVE BRAKING SYSTEM PROS AND CONS

Pros

Electricity generated from kinetic energy is transmitted back to power the train or to the overhead wire to power other trains

Cuts back on energy use by trains hence is more environmentally friendly

Cons

Less effective at low speeds

They occasionally malfunction because they have complex circuits so cannot be used as back up

HYDRAULIC BRAKING SYSTEM PROS AND CONS

Pros

Simple construction so is easy to manufacture maintain and repair

Low break wear rate so is cost effective

Equal and consistent braking action on all wheels

Cons

System could fail if there is a fluid leakage

Requires more maintenance compared to other systems

Air contamination affects the brakes efficiency and could cause brake failure

malfunction because they have complex circuits. For this reason they cannot be used as emergency brakes. In an electrical braking system, the braking force of the traction motor (generator) is transmitted to the wheels via gears. The generated electricity is adjusted to control braking force.

Mechanical Braking Systems

The basic braking devices used by mechanical braking systems are: wheel tread brakes (Fig. 8), axle-mounted disc brakes (Fig. 9), and wheel-mounted disc brakes (Fig. 10). All these mechanisms use an object (a brake shoe or lining) that applies friction to the disc. The basic braking devices used by mechanical braking systems are: wheel tread brakes (Fig. 8), axle-mounted disc brakes (Fig. 9), and wheel-mounted disc brakes (Fig. 10). All these mechanisms use an object (a brake shoe or lining) that applies friction to the disc. The applied pressure is adjusted to control the braking force. In the wheel-tread brake, the brake shoe applies friction to the wheel tread, creating a sliding effect. High-speed trains cannot use this type of brake, since doing so would damage the wheel tread. Instead, they use axle- or wheel-mounted disc brakes. Axle-mounted disc brakes are used on trailer bogies, because they have sufficient space to accommodate such a system. Wheel-mounted disc brakes are used on motor bogies that must accommodate the traction motor and have insufficient space for an axle-mounted brake.

Hydraulic Braking System

A hydraulic braking system works by converting mechanical force from the brake pedal into hydraulic pressure, which is transmitted through hydraulic lines to actuate brake calipers or wheel cylinders, creating friction on the brake discs or drums to slow down or stop the vehicle.
Hydraulic brake system uses a piston-cylinder structure and leverage to transmit the force of the hydraulic cylinder brake into the brake disc. The principle is often used by oil pressure driven piston motion transmission, transmission shift piston to tension brake system brake shoe soles applied to squeeze the brake shoe causing the disc brake stopping power cause arises.

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